

Wenhao Peng

RF Acoustic Wave Devices

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Biography

Wenhao Peng is an RF acoustic wave filter technology engineer at Skyworks Solutions, working across device modeling, filter design, and high-volume manufacturing of bulk acoustic wave resonators and filters. His work connects first-principles material and acoustic physics with practical device and filter design. In ferroelectric BAW technology, he developed a physics-based equivalent-circuit framework that bridges nonlinear ferroelectric thermodynamics and measured resonator behavior. In piezoelectric FBAR technology, he established an analytical relation between field-based and resonance-frequency-based definitions of electromechanical coupling, making modal field distributions directly useful for evaluating layer stacks and device performance. He is now extending modal methods to reduce the computational cost of finite-element acoustic simulation for production filter design. He also demonstrated the first FBAR operating above 50 GHz using a periodically poled AlN/ScAlN/AlN trilayer. He received the Ph.D. degree in electrical and computer engineering from the University of Michigan, Ann Arbor, in 2026.

Selected Contributions

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| Modal analysis | Proved the analytical relation between the two standard definitions of electromechanical coupling (k_t^2): the Berlincourt form computed from acoustic field distributions, and the form determined from the series and parallel resonance frequencies. Derived by modal analysis of the open-circuit eigenmodes of an FBAR, answering a question the literature had left open. The result makes acoustic field plots quantitatively interpretable, showing what a given mode contributes to resonator coupling, and converts layer stack design from iterative optimization against simulated admittance into direct evaluation. Quoted in both mm-wave FBAR papers, with the derivation presented at IMS 2024 and IMS 2025; now extended to cut finite element simulation cost in production filter design (see Experience). |
| Finite element methods | Adapted interpolation methods from finite element analysis into the thermodynamic modeling of ferroelectrics, bridging a bulk thermodynamic description and a compact circuit representation. Produced the first equivalent-circuit model of ferroelectric-driven BAW resonators derived from first principles rather than empirical fitting, reproducing measured response across DC bias. First author, IEEE TMTT 2022 and IMS 2021. |
| mm-Wave scaling | First FBAR demonstrated above 50 GHz: a 56 GHz periodically poled AlN/ScAlN/AlN trilayer, extending BAW operation roughly five times beyond the prior state of the art below 10 GHz. A second-generation 36 GHz device applied the modal analysis result above to electrode and layer stack selection. First author on both (IMS 2024, IMS 2025). Built on co-demonstrated controlled ferroelectric polarization switching in ultrawide-bandgap AlN/ScAlN multilayers (Applied Physics Letters 2023). |
| Switchless frontends | Co-developed the first switchless quad-band filter bank built on ferroelectric BST FBARs. Related work produced the first intrinsically switchable balanced ferroelectric FBAR filter and the first RF acoustic phase inversion switch based on poled ferroelectrics. |

Experience

- Feb 2026–Present **RF Acoustic Staff Electrical Engineer**, *Skyworks Solutions, Inc.*, Irvine, CA.
- Serve as technical bridge across device modeling, filter design, and high-volume manufacturing, applying equivalent-circuit and multiphysics modeling to guide design decisions and troubleshoot performance across the design-to-production pipeline.
 - Contribute to resonator stack design and optimization for higher-frequency cellular bands, balancing performance, tunability, and manufacturability requirements.
 - Extending the modal analysis methods developed in thesis work to reduce the computational cost of finite element acoustic simulation, with the goal of making mask-level response prediction deployable inside the filter design flow, where full finite element simulation had previously been impractical on runtime and cost grounds (ongoing R&D).
 - Work with modeling, process, and production teams to align design targets with fab capability and yield.
- Sep 2019–Dec 2025 **Graduate Student Research Assistant**, *University of Michigan*, Ann Arbor, MI.
Advisor: Prof. Amir Mortazawi
- 2025 *mm-Wave acoustic resonators based on periodically poled III-nitride thin films*
- Demonstrated the first mm-wave FBARs using AlN/ScAlN/AlN periodically poled trilayers, including the first device of this class operating above 50 GHz (56 GHz), followed by a second-generation 36 GHz resonator with a redesigned electrode and layer stack. On-wafer characterization in the dissertation gives $k_t^2 = 1.7\%$, $Q_m = 53$ at 56 GHz and $k_t^2 = 2.9\%$, $Q_m = 102$ at 36 GHz.
 - Proved the Berlincourt-to-resonance-frequency coupling relation described above, as the theoretical groundwork for the device work, replacing iterative optimization against admittance responses with direct evaluation.
 - Advanced the nanofabrication and pulsed-field polarization-switching processes, including current-response analysis.
 - Performed mm-wave characterization to 67 GHz with on-wafer de-embedding structures validated in ADS Momentum and Ansys HFSS.
- Reconfigurable RF frontends based on intrinsically switchable BST acoustic resonators*
- Built the first equivalent-circuit model of ferroelectric-driven BAW resonators grounded in ferroelectric thermodynamics, adapting interpolation methods from finite element analysis to bridge the bulk thermodynamic description and a compact circuit representation, and capturing the underlying physics rather than fitting measured response. Validated by direct overlay of simulated and measured response across DC bias.
 - Contributed to the first switchless quad-band BST FBAR filter bank and the first intrinsically switchable balanced ferroelectric FBAR filter at 2 GHz.
 - Contributed to the first RF acoustic phase inversion switch based on poled-ferroelectric stacked crystal filters.
 - Synthesized filters by the image parameter method, applying Mason and mBVD resonator models, then carried them through full cleanroom fabrication and DC-bias-dependent characterization.

Education

- 2026 **Ph.D., Electrical and Computer Engineering**, *University of Michigan*, Ann Arbor, MI.
Dissertation: *Layered Ferroelectric Millimeter-Wave BAW Resonators: Modeling and Validation based on On-Wafer Characterization*. Advisor: Prof. Amir Mortazawi.
- 2024 **M.S.E., Mechanical Engineering**, *University of Michigan*, Ann Arbor, MI.
Dynamics and vibrations
- 2019 **M.S., Electrical and Computer Engineering**, *University of Michigan*, Ann Arbor, MI.
Integrated circuits and VLSI

- 2018 **B.S.E., Electrical Engineering**, *University of Michigan*, Ann Arbor, MI.
Summa Cum Laude
- 2018 **B.S.E., Electrical and Computer Engineering**, *Shanghai Jiao Tong University*,
Shanghai, China.
Dual degree with U-M

Technical Skills

Devices & materials	BAW/FBAR resonator and filter design; thin-film piezoelectrics and ferroelectrics (AlN, ScAlN, BST); periodically poled III-nitride multilayers; MEMS; RF frontend architectures
Modeling	Mason and mBVD equivalent circuits; thermodynamic modeling of ferroelectric nonlinearity; modal analysis; image-parameter filter synthesis; multiphysics FEM (COMSOL); EM co-simulation (ADS Momentum, Ansys HFSS); structural vibration analysis
Fabrication	Cleanroom process integration; photolithography; physical vapor deposition; wet/dry/plasma etching; thermal oxidation; device processing compatible with MBE-grown films; pulsed-field poling
Measurement	On-wafer S-parameter characterization to 67 GHz; GSG probing; VNA calibration and de-embedding; DC-bias-dependent RF characterization; material parameter extraction
Tools	ADS, COMSOL Multiphysics, Ansys HFSS, MATLAB, Python, Cadence, LLM

Journal Publications

- 2025 H. Desai, **W. Peng**, and A. Mortazawi, "Single-Pole Single-Throw RF Acoustic Phase Inversion Switch Leveraging Poled Ferroelectrics," *IEEE Trans. Microw. Theory Techn.*, vol. 73, no. 1, pp. 6–13, 2025.
- 2023 D. Wang, P. Wang, S. Mondal, J. Liu, M. Hu, M. He, S. Nam, **W. Peng**, S. Yang, D. Wang, Y. Xiao, Y. Wu, A. Mortazawi, and Z. Mi, "Controlled Ferroelectric Switching in Ultrawide Bandgap AlN/ScAlN Multilayers," *Appl. Phys. Lett.*, vol. 123, no. 10, p. 103506, 2023.
- 2023 S. Nam, **W. Peng**, P. Wang, D. Wang, Z. Mi, and A. Mortazawi, "A mm-Wave Trilayer AlN/ScAlN/AlN Higher Order Mode FBAR," *IEEE Microw. Wireless Technol. Lett.*, vol. 33, no. 6, pp. 803–806, 2023.
- 2022 **W. Peng**, M. Z. Koohi, S. Nam, and A. Mortazawi, "Phenomenological Circuit Modeling of Ferroelectric-Driven Bulk Acoustic Wave Resonators," *IEEE Trans. Microw. Theory Techn.*, vol. 70, no. 1, pp. 919–925, 2022.
- 2021 S. Nam, M. Z. Koohi, **W. Peng**, and A. Mortazawi, "A Switchless Quad Band Filter Bank Based on Ferroelectric BST FBARs," *IEEE Microw. Wireless Compon. Lett.*, vol. 31, no. 6, pp. 662–665, 2021.

Conference Publications

- 2025 **W. Peng**, S. Nam, D. Wang, Z. Mi, and A. Mortazawi, “A 36 GHz Trilayer AlN/ScAlN/AlN Periodically Poled FBAR,” *IEEE/MTT-S International Microwave Symposium (IMS)*, San Francisco, CA, 2025, pp. 786–789. *(presented by author)*
- 2024 **W. Peng**, S. Nam, D. Wang, Z. Mi, and A. Mortazawi, “A 56 GHz Trilayer AlN/ScAlN/AlN Periodically Poled FBAR,” *IEEE/MTT-S International Microwave Symposium (IMS)*, Washington, DC, 2024, pp. 150–153. *(presented by author)*
- 2021 **W. Peng**, M. Z. Koohi, S. Nam, and A. Mortazawi, “Physics Based Modeling of Electrostriction Based BAW Resonators,” *IEEE MTT-S International Microwave Symposium (IMS)*, Atlanta, GA, 2021, pp. 214–217. *(presented by author)*
- 2020 M. Z. Koohi, **W. Peng**, and A. Mortazawi, “An Intrinsically Switchable Balanced Ferroelectric FBAR Filter at 2 GHz,” *IEEE/MTT-S International Microwave Symposium (IMS)*, 2020, pp. 131–134.
- 2018 Y. Dai, **W. Peng**, Y. Wang, L.-X. Chuo, K. Suri, H. Zheng, D. Wentzloff, and H.-S. Kim, “Implementation and Evaluation of Bi-Directional WiFi Back-Channel Communication,” *IEEE 29th Annual International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC)*, 2018, pp. 1–7.

Peer Review

- 40 reviews **IEEE:** J. Microelectromechanical Systems (7), Microwave and Wireless Technology Letters (6), Electron Device Letters (4), J. Microwaves (1). **APS:** Physical Review Letters (5), Physical Review Applied (2), Physical Review Materials (2), Physical Review B (1). **Other:** Micromachines (7), Photonics (2), Microwave (MDPI) (2), Progress in Quantum Electronics (1).

Selected Graduate Coursework

Wave Propagation in Elastic Solids; Advanced Vibrations of Structures; Advanced Engineering Acoustics; Theory of Solid Continua; Math Methods in ME; Advanced Solid State Microwave Circuits; Electromagnetics Theory; Microwave Circuits; Introduction to MEMS; Monolithic Amplifier Circuits; VLSI Design I/II